Method of Differentiation: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. First principle and notation

Formulas

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

KEY POINTS

- $f'(x)$ = instantaneous rate of change; slope of the tangent
- $\frac{dy}{dx} = Dy = f'(x) = y_1 = y'$; a dot means $\frac{d}{dt}$: $\dot{S} = \frac{dS}{dt}$
- $\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = f'(a)$: such limits are derivatives in disguise

2. Standard derivatives

Formulas

$$(x^n)' = nx^{n-1},\ (e^x)' = e^x,\ (a^x)' = a^x \ln a,\ (\ln x)' = \frac{1}{x},\ (\log_a x)' = \frac{1}{x \ln a}$$

$$(\sin x)' = \cos x,\ (\cos x)' = -\sin x,\ (\tan x)' = \sec^2 x,\ (\cot x)' = -\operatorname{cosec}^2 x$$

$$(\sec x)' = \sec x \tan x,\ (\operatorname{cosec} x)' = -\operatorname{cosec} x \cot x$$

$$(\sin^{-1} x)' = -(\cos^{-1} x)' = \frac{1}{\sqrt{1-x^2}},\ (\tan^{-1} x)' = -(\cot^{-1} x)' = \frac{1}{1+x^2}$$

$$(\sec^{-1} x)' = -(\operatorname{cosec}^{-1} x)' = \frac{1}{|x|\sqrt{x^2-1}}$$

KEY POINTS

- Co-functions (cos, cot, cosec and their inverses) carry a minus sign
- Simplify first: $e^{\ln g} = g$, $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, $2^x e^x = (2e)^x$

3. Sum, product and quotient rules

Formulas

$$(fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$$

$$(fgh)' = f'gh + fg'h + fgh' = \frac{1}{2}\left[(fg)'h + (gh)'f + (hf)'g\right]$$

KEY POINTS

- $(f \pm g)' = f' \pm g'$, $(kf)' = kf'$
- Before the quotient rule: factorise, cancel, use identities
- Long sums/products: use sigma or log form, e.g. $f = 1 + x + \cdots + x^{100} \Rightarrow f'(1) = 5050$

4. Chain rule and composite functions

Formulas

$$\frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x), \qquad \frac{dy}{dx} = \frac{1}{dx/dy}$$

STEPS

1. Name the outer function and the inner function
2. Differentiate outside to inside, one layer at a time
3. Multiply all the factors; put the point only at the end

Watch out

$f'(g(x))$ means f' evaluated at $g(x)$, not $\frac{d}{dx} f(g(x))$

5. Logarithmic differentiation

Formulas

$$\frac{d}{dx} f^{\,g} = f^{\,g}\left[g' \ln f + \frac{g f'}{f}\right]$$

KEY POINTS

- Use for long products/quotients and for $[f(x)]^{g(x)}$
- Shortcut: (power rule, exponent fixed) + (exponential rule, base fixed)
- Sum of such terms ($x^x + (\sin x)^x$): differentiate each term separately

STEPS

1. $\ln y =$ sum of logs
2. Differentiate: $\frac{y'}{y} = \ldots$
3. Multiply by y

Watch out

Do not take ln of a sum term by term

6. Parametric and one function w.r.t. another

Formulas

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \qquad \frac{d\,f(x)}{d\,g(x)} = \frac{f'(x)}{g'(x)}$$

KEY POINTS

- f w.r.t. g is a parametric derivative with x as the parameter
- ITFs: substitute $x = \tan\theta, \sin\theta, \cos\theta$ to simplify both, then divide
- The simplified form depends on the interval of x: check the range of θ

7. Implicit functions

Formulas

$$f(x,y) = 0 \Rightarrow \frac{dy}{dx} = -\frac{f_x}{f_y}$$

STEPS

1. Differentiate every term; $\frac{d}{dx}\phi(y) = \phi'(y)\,y'$
2. Collect y' terms and solve; put the point at the end

KEY POINTS

- Simplify first: $e^{x-y} = x^y \Rightarrow y = \frac{x}{1+\ln x}$; $x\sqrt{1+y} + y\sqrt{1+x} = 0 \Rightarrow y = -\frac{x}{1+x}$
- Homogeneous equation $= 0$ (e.g. $ax^2 + 2hxy + by^2 = 0$): $y' = \frac{y}{x}$, $y'' = 0$
- Infinite radicals/powers: $y = \sqrt{x+y}$, $y = x^y$ - replace the repeated part by y

8. Inverse function

Formulas

$$g = f^{-1}: \; g'(b) = \frac{1}{f'(a)}, \quad g''(b) = -\frac{f''(a)}{(f'(a))^3}, \quad f(a) = b$$

STEPS

1. Solve $f(a) = b$ (by inspection: try $0, \pm 1, \pm 2$)
2. $g'(b) = 1/f'(a)$

Watch out

$g'(b) \neq \frac{1}{f'(b)}$: use the pre-image a, not b

9. Higher order derivatives

Formulas

$$y_2 = \frac{d}{dx}(y_1); \quad \left.\frac{d^2y}{dx^2}\right|_{\text{param}} = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^3}; \quad \frac{d^2x}{dy^2} = -\frac{y_2}{y_1^3}$$

KEY POINTS

- Notation: $y_2 = y'' = f''(x) = D^2y = \frac{d^2y}{dx^2}$
- y' = slope, y'' = concavity
- To prove $Py_2 + Qy_1 + Ry = 0$: square / clear fractions first, then differentiate again
- n-th derivative: find y_1, y_2, y_3 and spot the pattern (sin: shift by $\frac{\pi}{2}$ each time)

Watch out

$\frac{d^2y}{dx^2} \neq \frac{d^2y/dt^2}{d^2x/dt^2}$ and $\frac{d^2x}{dy^2} \neq \frac{1}{d^2y/dx^2}$

10. Analysis of ITF

Formulas

$$\sin^{-1}\frac{2x}{1+x^2} = 2\tan^{-1}x \ (|x| \leq 1), \ \pi - 2\tan^{-1}x \ (x > 1), \ -\pi - 2\tan^{-1}x \ (x < -1)$$

$$\cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1}x \ (x \geq 0), \ -2\tan^{-1}x \ (x < 0)$$

STEPS

1. Domain, range; piecewise form by substitution
2. Limits at $\pm\infty$ and at the joins
3. f': monotonic intervals and non-derivable points
4. f'': concavity; then draw

11. Identities, expansions, divisibility

Formulas

$$a_r = \frac{f^{(r)}(0)}{r!}; \qquad (x-a)^2 \mid f \iff f(a) = f'(a) = 0$$

KEY POINTS

- Only an identity may be differentiated on both sides
- Differentiate $(1+x)^n = \sum \binom{n}{r} x^r$ to get $\sum r\binom{n}{r} = n2^{n-1}$ and similar sums
- Repeated root a of $f = 0$ is also a root of $f' = 0$

12. Differentiation of determinants

KEY POINTS

- Differentiate one row (or column) at a time and add the determinants
- Constant rows give zero terms; two equal rows give zero
- For $\Delta'(a)$: differentiate, then put $x = a$

13. L'Hospital rule

Formulas

$$\lim\frac{f}{g} = \lim\frac{f'}{g'} \ \left(\frac{0}{0}, \frac{\infty}{\infty}\right); \qquad \lim f^{\,g} = e^{\lim g \ln f}$$

KEY POINTS

- Point a finite or $\pm\infty$; f, g differentiable near a
- Can be repeated while the form stays $\frac{0}{0}$ or $\frac{\infty}{\infty}$
- $0 \cdot \infty \to \frac{0}{1/\infty}$; $\infty - \infty \to$ single fraction; $0^0, \infty^0, 1^\infty \to$ take ln

Watch out

Differentiate Nr and Dr separately - not the quotient rule
Check the form before each application

14. Differentiation of integrals

Formulas

$$\frac{d}{dx}\int f\,dx = f; \qquad \frac{d}{dx}\int_{g(x)}^{h(x)} f(t)dt = f(h)h' - f(g)g'$$

KEY POINTS

- Take x out of the integrand before applying the rule
- Limits of the form $\frac{\int_0^x f}{x^n}$: L'Hospital + Leibniz

15. Common mistakes checklist

STEPS

1. Simplify (identities, cancelling, ITF substitution) before differentiating
2. Chain rule factor $g'(x)$ not forgotten
3. $(f^g)'$ has two parts: power part + exponential part
4. Inverse: $g'(b) = 1/f'(a)$ with $f(a) = b$
5. Parametric y'': divide $\frac{d}{dt}(y')$ by $\dot{x}$
6. ITF piecewise forms change at $x = \pm 1$ or 0
7. L'Hospital only in $\frac{0}{0}$, $\frac{\infty}{\infty}$